Year 1 AS Pure Maths

Chapter 13

Integration

Practice Booklet

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13.1 Integrating x^n

Practice Book, printed page 104.

1. Find $f(x)$ when $f'(x)$ equals:

a. $9x^2$

b. $12x^{-3}$

c. $6x^{5/2}$

d. 25

Hint

If $f'(x) = x^n$, then

$$f(x) = \frac{1}{n+1}x^{n+1} + c,\ n \neq -1$$

2. Find y when $\frac{dy}{dx}$ equals:

a. $\sqrt{x}$

b. $\frac{3}{\sqrt[3]{x}}$

c. $5\sqrt[4]{x}$

d. $-\frac{1}{6\sqrt{x}}$

Hint

First write each term in the form kx^n.
If $\frac{dy}{dx} = kx^n$, then $y = \frac{k}{n+1}x^{n+1} + c$, $n \neq -1$

3. Find $f(x)$ when $f'(x)$ is given by each of the following expressions, giving your coefficients as simplified fractions.

a. $x^4 + x^3 - 5x + 6$

b. $4x^2 + \frac{1}{3}x^{-2} - 5x^{-3}$

c. $3x^{-1/4} - 2x^{-3/4}$

Hint

Apply the rule of integration to each term in the expression and add c.

4. $f'(x) = (4x + 3)^2$

a. Write $f'(x)$ in the form $ax^2 + bx + c$.

b. Hence find $f(x)$.

Hint

First expand the brackets, then apply the rule of integration to each term in the expression and add the unknown constant.

5. E/P Find $f(x)$ given $f'(x) = (3x - 5)^2$. (4 marks)

6. E/P Find y given $\frac{dy}{dx} = (1 - 3x)^2$. (4 marks)

7. E/P $h'(x) = (p - 5x)^4$

a. The coefficient of the x^3 term in the expansion of $h'(x)$ is -1000.
Find the value of the constant p. (4 marks)

b. Hence, or otherwise, find $h(x)$. (2 marks)

8. E/P $f'(x) = (4 - 2\sqrt{x})^2$

a. Write $f'(x)$ in the form $a + b\sqrt{x} + cx$, where a, b and c are constants to be found. (2 marks)

b. Hence find $f(x)$. (3 marks)

9. E/P $g'(x) = (3 - 2x)^8$

a. Find the first three terms, in ascending powers of x, if the binomial expansion of $g'(x)$, giving each term in its simplest form. (4 marks)

b. Assuming x is small and higher powers can be ignored, use your answer to part **a** to find an approximation to $g(x)$ in the form $px + qx^2 + rx^3 + c$. (1 mark)

13.2 Indefinite integrals

Practice Book, printed page 105.

1. Find the following integrals:

a. $\int x^4 \, dx$

b. $\int -7x^6 \, dx$

c. $\int 4x^{-2} \, dx$

Hint

$\int kx^n \, dx = \frac{k}{n+1}x^{n+1} + c$, $n \neq -1$

2. Find the following integrals:

a. $\int (5x^2 + 9x^{-3}) \, dx$

b. $\int (\frac{3}{4}x^5 - 4x^{-5} + 8x) \, dx$

c. $\int (4x^{3/2} - 10x^{-3/2}) \, dx$

Hint

Integrate each term separately:

$$\int (f(x) + g(x)) \, dx = \int f(x) \, dx + \int g(x) \, dx$$

3. Find the following integrals:

a. $\int (6 - 5\sqrt{x})^2 \, dx$

b. $\int \frac{(3x-2)^2}{\sqrt{x}} \, dx$

c. $\int \left(\frac{5-7x^3+4x^{-3/2}}{x^2}\right) \, dx$

Hint

First expand the brackets or divide so each term is in the form kx^n, then integrate.

4. Find the following integrals:

a. $\int \left(\frac{P}{\sqrt{x}} - Rx^{3/2}\right) dx$

b. $\int \left(\frac{A}{x^3} + \frac{B}{x^5}\right) dx$

c. $\int (M\sqrt[3]{x} - Nx^{3/4})\, dx$

Hint

Treat letters other than x as constants.

5. E/P Given that $y^{1/2} = 6x^{1/5} - 5$, find $\int y\, dx$ (3 marks)

6. E Given that $g(x) = 3x^3 - 7\sqrt{x} + \frac{4}{\sqrt[3]{x}} - \frac{2}{x^2}$, find $\int g(x)\, dx$ (4 marks)

7. E/P $\int \left(abx^2 + \frac{4b}{x^5}\right) dx = 2x^3 - \frac{2}{x^4} + c$, where a and b are real constants.
Find the values of a and b. (4 marks)

8. E/P $g(x) = (1 + 2x)(4 - 3x)^8$
Given that x is small, and so terms in x^3 and higher powers of x can be ignored,

a. find an approximation for $g(x)$ in the form $P + Qx + Rx^2$ (4 marks)

b. find an approximation for $\int g(x)\, dx$. (3 marks)

13.3 Finding functions

Practice Book, printed page 105–106.

1. Find the equation of the curve with the given derivative of y with respect to x that passes through the given point:

a. $\frac{dy}{dx} = 4x^3 - 10x^2$; point $(3, 6)$

b. $\frac{dy}{dx} = \frac{5}{\sqrt{x}} + \frac{1}{2}x^2$; point $(16, 65)$

Hint

Apply the rule of integration and then substitute the given values of x and y to find the constant c.

2. Find the equation of the curve with the given derivative of y with respect to x that passes through the given point:

a. $\frac{dy}{dx} = \frac{3-x^2}{x^4}$; point $(-1, 4)$

b. $\frac{dy}{dx} = (x - 5)^2$; point $(2, 18)$

Hint

Begin by writing each term in the form ax^n, then integrate and finally substitute to find the value of the constant c.

3. The gradient of a curve, with equation $y = f(x)$, is given by $f'(x) = \frac{8}{\sqrt{x}} - 4\sqrt{x}$
The curve passes through the point $(9, -30)$.
Find the equation of the curve in the form $y = f(x)$.

4. E $\frac{dy}{dx} = 10 - 6x - 4x^2$. Given that $y = -15$ when $x = 3$, find y in terms of x. (4 marks)

5. E $f'(x) = \frac{5\sqrt{x}-3x^{3/2}}{x^2}$
The curve with equation $y = f(x)$ passes through the point $(4, 7)$. Find $f(x)$. (5 marks)

6. E The gradient of a curve, with equation $y = f(x)$, is given by $f'(x) = 6x - \frac{4}{\sqrt[3]{x}}$
The curve passes through the point $(8, 112)$.
Find the equation of the curve in the form $y = f(x)$. (4 marks)

7. E/P $p'(x) = \frac{(1-5x)^3}{\sqrt{x^3}}$

a. Show that $p'(x) = ax^{-3/2} + bx^{-1/2} + cx^{1/2} + dx^{3/2}$ where a, b, c and d are constants to be found. (4 marks)

b. Given that the point $(1, -24)$ lies on the curve with equation $y = p(x)$, find $p(x)$. (3 marks)

8. E/P Given that $h'(x) = \frac{3}{2}x^2 - \frac{1}{4\sqrt{x}}$ and $h(4) = -10$, find the exact value of $h(2)$. (4 marks)

13.4 Definite integrals

Practice Book, printed page 106–107.

1. Evaluate the following definite integrals.

a. $\int_2^{10} x^2\,dx$

b. $\int_1^8 6\sqrt[3]{x}\,dx$

c. $\int_2^4 \frac{4}{x^3}\,dx$

d. $\int_4^9 \frac{5}{\sqrt{x}}\,dx$

Hint

If $f'(x)$ is the derivative of $f(x)$ for all values of x in the interval $[a, b]$ then the definite integral is defined as

$$\int_a^b f'(x)\,dx = [f(x)]_a^b = f(b) - f(a)$$

2. Evaluate the following definite integrals.

a. $\int_1^2 \left(18x - \frac{4}{x^3}\right)\,dx$

b. $\int_1^4 \frac{1}{\sqrt{x}}(14x^3 - 5x^2)\,dx$

c. $\int_1^{36} \left(2\sqrt{x} - \frac{6}{x}\right)^2\,dx$

3. Use calculus to find the value of $\int_1^2 \left(\frac{5}{x^2} - \frac{3}{x^4}\right)\,dx$

Hint

This question says 'use calculus' so you need to use integration and show the integrated function. Do not use a numerical integration function on your calculator.

4. E Use calculus to find the value of $\int_1^{16} \left(\frac{2}{x^2} + 4\sqrt{x}\right)\,dx$ (4 marks)

5. E Use calculus to find the value of $\int_1^4 \left(\frac{8x^3}{3} - \frac{6}{x^2} - \frac{2}{\sqrt{x}}\right)\,dx$ (4 marks)

6. E/P Given that $\int_4^k 9\sqrt{x}\,dx = 702$, find the value of k. (4 marks)

7. E $y^{1/2} = 2x^{1/3} - \frac{1}{2}x^{-1/3}$

a. Show that $y = ax^{2/3} + bx^{-2/3} + c$, where a, b and c are constants to be found. (2 marks)

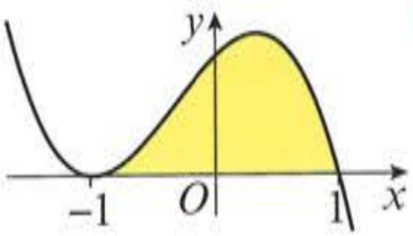
b. Find $\int_1^8 y\,dx$ (3 marks)

8. E/P Given that $\int_4^9 a\sqrt{x}\,dx = -114$, find the value of a. (3 marks)

13.5 Areas under curves

Practice Book, printed page 107–108.

1. The diagram shows part of the curve C with equation $y = -x^3 - x^2 + x + 1$.



Find the area of the shaded region.

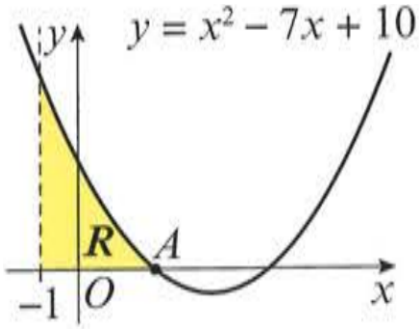
Hint

The area between a positive curve, the x-axis and the lines $x = a$ and $x = b$ is given by

$$\text{Area} = \int_a^b y\,dx$$

where $y = f(x)$ is the equation of the curve.

2. The diagram shows a sketch of the curve with equation $y = x^2 - 7x + 10$.



Find:

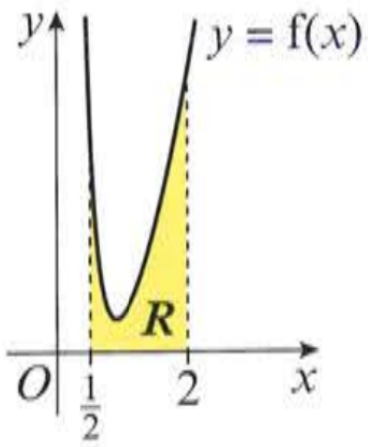
a. the coordinates of point A

b. the area of the shaded region R.

Hint

First solve $x^2 - 7x + 10 = 0$. The x-coordinate of the point A will be the smaller of the two answers.

3. $f(x) = 5x^2 + \frac{2}{x^3} - 5$, $x > 0$.
The diagram shows the curve with equation $y = f(x)$.

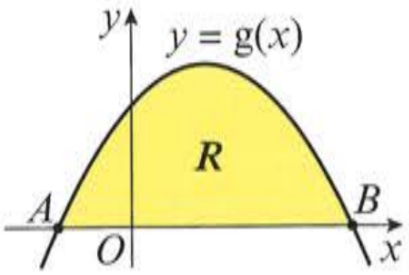


The region R is bounded by the curve, the x-axis and the lines $x = \frac{1}{2}$ and $x = 2$. Use calculus to find the exact area of R.

Hint

The question says 'use calculus', so you need to show the integrated function in your working.

4. E The diagram shows the curve with equation $y = g(x)$, where $g(x) = 12 + 4x - x^2$, $x \geq 0$.

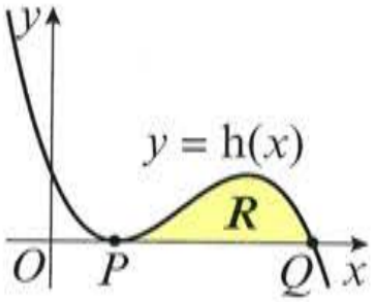


The curve meets the x-axis at the points A and B. Find:

a. the coordinates of points A and B (2 marks)

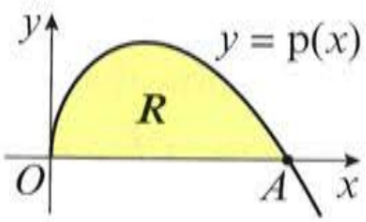
b. the exact area of region R. (4 marks)

5. E/P $h(x) = -(x-1)^2(x-4)$. The diagram shows the curve with equation $y = h(x)$.



The curve meets the x-axis at the points P and Q.
Use calculus to find the exact area of region R. (6 marks)

6. E $p(x) = 8\sqrt{x} - x^2$. The diagram shows the curve with equation $y = p(x)$.

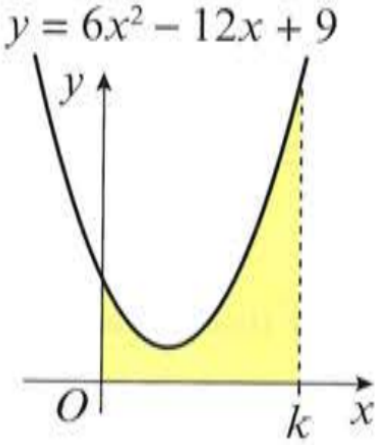


The curve meets the x-axis at the origin and at the point A.
Find:

a. the coordinates of point A (2 marks)

b. the exact area of region R. (4 marks)

7. E/P The shaded region is bounded by the curve with equation $y = 6x^2 - 12x + 9$, the x- and y-axes and the line $x = k$. Given that the area of the shaded region is 27, find the value of k.

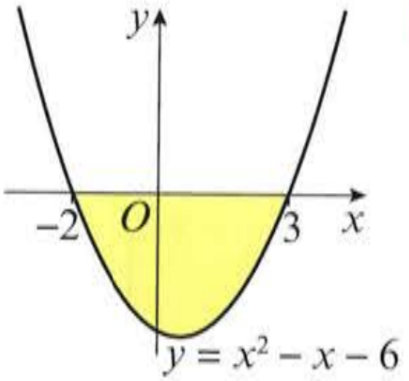


(6 marks)

13.6 Areas under the x-axis

Practice Book, printed page 108–109.

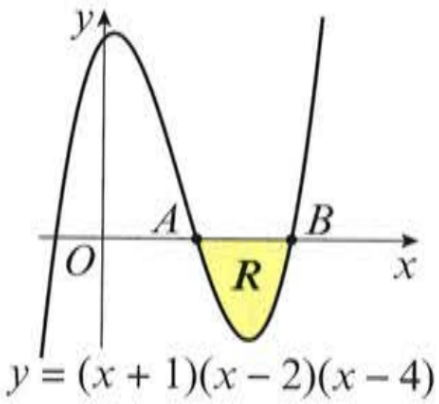
1. Find the area of the finite region bounded by the curve $y = x^2 - x - 6$ and the x-axis.



Hint

Find $\int_{-2}^{3}(x^2 - x - 6)\,dx$. The value of the definite integral will be negative because the area is below the x-axis, but you should state the positive quantity as the final answer.

2. The graph shows a sketch of part of the curve C with equation $y = (x+1)(x-2)(x-4)$. The curve cuts the x-axis at $(-1, 0)$ and at the points A and B.



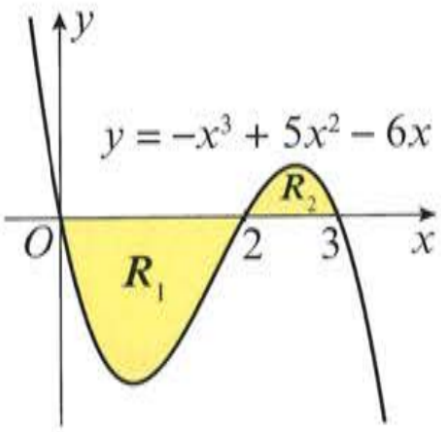
a. Write down the x-coordinates of the points A and B.

b. Find the area of the shaded region R.

Hint

The limits of the integral will be the x-coordinates of points A and B.

3. The diagram shows the region bounded by the curve $y = -x^3 + 5x^2 - 6x$ and the x-axis.



Find:

a. the area of region R_1

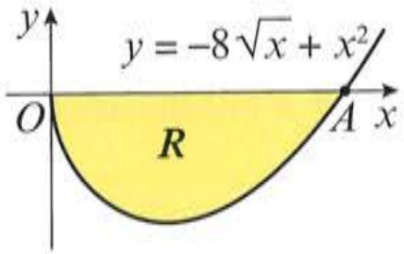
b. the area of region R_2

c. the total area of the shaded region.

Hint

To find the total shaded area, find the areas of the regions above the x-axis and below the x-axis separately.

4. E The graph shows a sketch of part of the curve C with equation $y = -8\sqrt{x} + x^2$

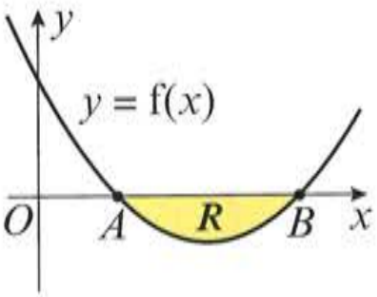


The shaded region R is bounded by the curve and the x-axis.

a. Find the coordinates of the point A. (1 mark)

b. Use calculus to find the exact area of the shaded region R. (5 marks)

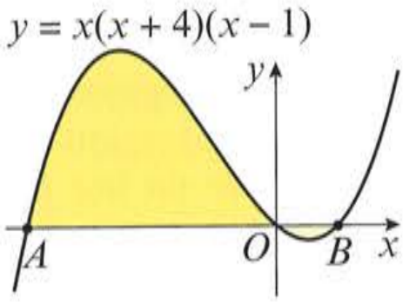
5. E $f(x) = \frac{1}{2}x^2 - \frac{13}{2}x + 15$. The graph shows a sketch of part of the curve with equation $y = f(x)$.



a. Find the coordinates of points A and B. (2 marks)

b. Find the exact area of the shaded region R. (4 marks)

6. E/P The graph shows a sketch of part of the curve C with equation $y = x(x+4)(x-1)$

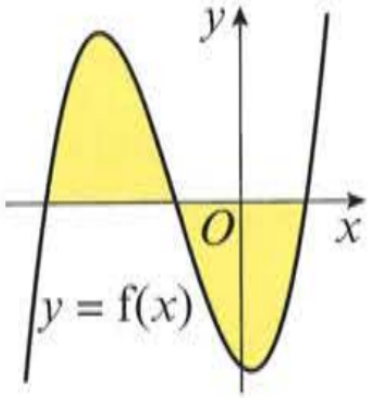


The curve cuts the x-axis at the origin O and at the points A and B.

a. Write down the coordinates of points A and B. (1 mark)

b. Find the total area of the shaded region. (7 marks)

7. E/P $f(x) = x^3 + 6x^2 - 4x - 24$. The graph shows a sketch of part of the curve with equation $y = f(x)$.



a. Show that $f(2) = 0$. (1 mark)

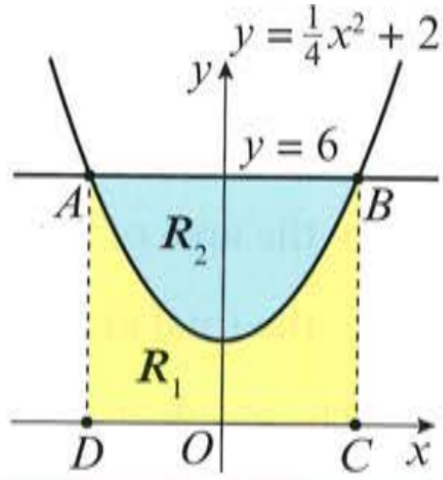
b. Hence, or otherwise, factorise $f(x)$ completely. (2 marks)

c. Find the total shaded area shown in the sketch. (5 marks)

13.7 Areas between curves and lines

Practice Book, printed page 110–111.

1. The diagram shows part of the curve with equation $y = \frac{1}{4}x^2 + 2$ and the line with equation $y = 6$. The line cuts the curve at the points A and B.



a. Find the coordinates of points A and B.

b. Use integration to find the shaded region R_1.

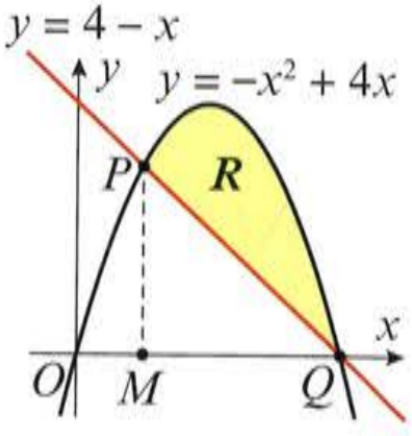
c. Find the area of the rectangle $ABCD$.

d. Find the area of the shaded region R_2.

Hint

Area of R_2 = area of rectangle−area of R_1

2. The diagram shows part of the curve with equation $y = -x^2+4x$ and the line with equation $y = 4-x$. The line cuts the curve at the points P and Q.



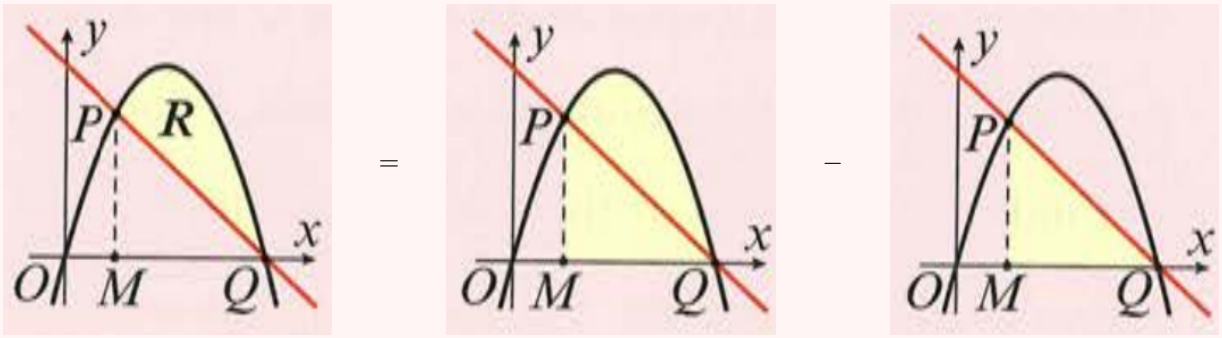
a. Find the coordinates of the points P and Q.

b. Use integration to find the shaded region bounded by the curve, the line PM and the x-axis.

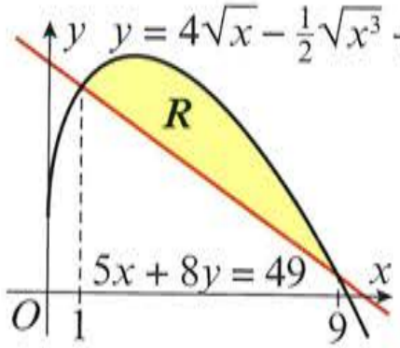
c. Find the area of the triangle PQM.

d. Find the area of the shaded region R.

Hint



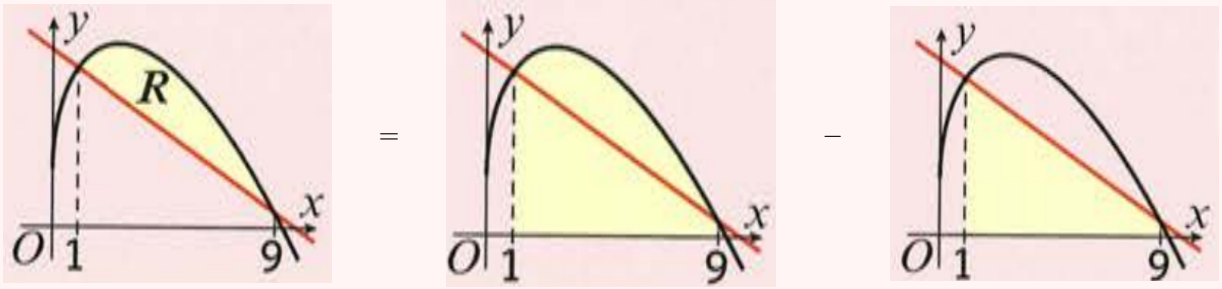
3. The diagram shows part of the curve with equation $y = 4\sqrt{x} - \frac{1}{2}\sqrt{x^3} + 2$ and the line with equation $5x + 8y = 49$.



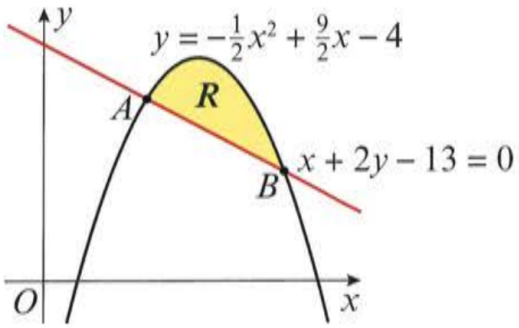
a. Use integration to find the shaded region bounded by the curve, the lines $x = 1$, $x = 9$ and the x-axis.

b. Find the shaded region R.

Hint



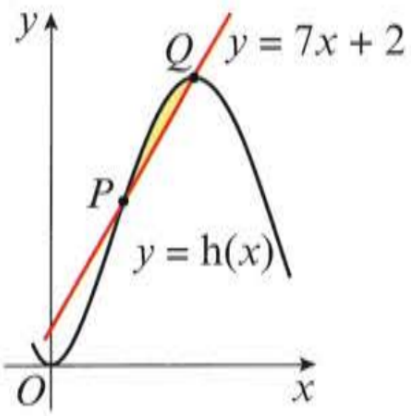
4. E/P The shaded region R is bounded by the curve with equation $y = -\frac{1}{2}x^2 + \frac{9}{2}x - 4$ and the line with equation $x + 2y - 13 = 0$ as shown in the diagram. The line intersects the curve at the points A and B.



a. Find the coordinates of the points A and B. (3 marks)

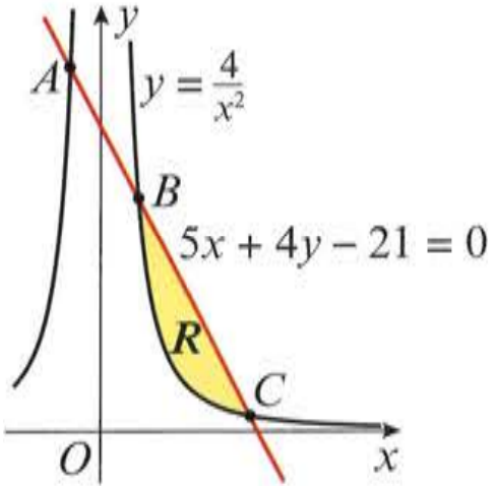
b. Find the area of the shaded region R. (5 marks)

5. E/P $h(x) = x^2(x-4)^2$. The graph shows part of the curve with equation $y = h(x)$.



The straight line $y = 7x + 2$ intersects the curve at the points P and Q with x-coordinates 1 and 2 respectively. Find the area of the finite region bounded by the curve and the segment PG. (7 marks)

6. E/P The line with equation $5x + 4y - 21 = 0$ crosses the curve with equation $y = \frac{4}{x^2}$ at the points A, B and C.



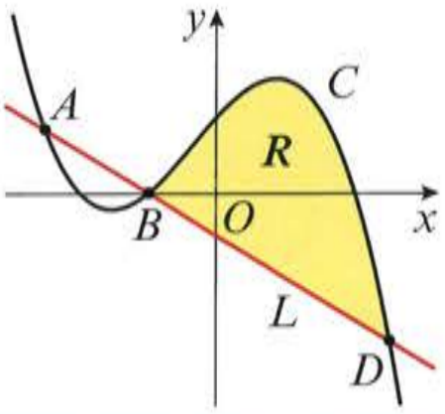
a. Show that at points A, B and C, $5x^3 - 21x^2 + 16 = 0$. (3 marks)

Given that the x-coordinate of B is 1, find:

b. the coordinates of the point C (2 marks)

c. the area of the shaded region R. (7 marks)

7. E/P The line L with equation $3x+y+6=0$ crosses the curve C with equation $3y=-x^3-2x^2+16x+32$ at the points A, B and D.



a. Show that at points A, B and C, $x^3+2x^2-25x-50=0$. (2 marks)

Given that B has coordinates $(-2,0)$, find:

b. the coordinates of the point D (2 marks)

c. the area of the shaded region R. (7 marks)

Problem set A

Bronze

$f'(x) = \frac{-5x^2+7x^{2/3}}{\sqrt{x}}$

a. Write $f'(x)$ in the form $ax^m + bx^n$, where a, b, m and n are rational numbers to be found. (2 marks)

b. Given that $f(1) = -10$, find an expression for $f(x)$. (4 marks)

Silver

$f'(x) = (1 - 2x^{1/3})^3$. Given that $f(8) = 24$, find $f(1)$. (6 marks)

Gold

$f'(x) = 3x^2 + px + \frac{8}{x^2}$. Given that $f(-4) = -85$ and $f(2) = 11$, find $f(x)$. (6 marks)

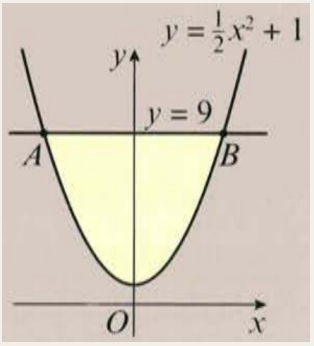
Problem set B

Bronze

The diagram shows part of the curve with equation $y = \frac{1}{2}x^2 + 1$ and the line with equation $y = 9$. The line cuts the curve at the points A and B.

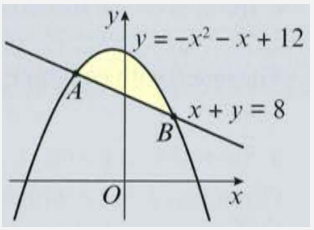
a. Find the coordinates of the points A and B. (2 marks)

b. Find the area of the finite region bounded by line AB and the curve. (6 marks)



Silver

The diagram shows part of the curve with equation $y = -x^2 - x + 12$ and the line with equation $x + y = 8$. The line cuts the curve at the points A and B. Find the area of the finite region bounded by line AB and the curve. (9 marks)

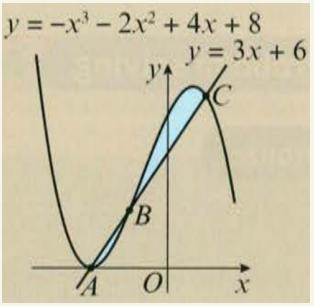


Gold

The diagram shows part of the curve with equation $y = -x^3 - 2x^2 + 4x + 8$ and the line with equation $y = 3x + 6$. The line cuts the curve at the points A, B and C.

a. Show that at points A, B and C, $x^3 + 2x^2 - x - 2 = 0$. (2 marks)

b. Given that A has coordinates $(-2, 0)$, find the total area of the region bounded by the curve and the line segment BC. (9 marks)



Selected exam question bank questions

Question 6

Given that $f(x) = 9\sqrt[3]{x^2} + \frac{4}{x^2} - 3x^3 + 2$, find $\int f(x)\,dx$. (4 marks)

Question 8

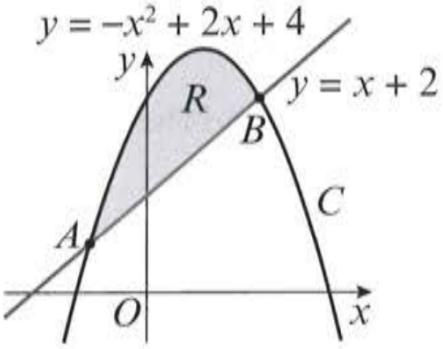
Find $\int_{1/2}^{1}(4-3x)^3\,dx$, leaving your answer as a fraction in its simplest terms. (5 marks)

Question 26

Given that $\int_9^k \frac{8}{\sqrt{x}}\,dx = 32$, find the value of k. (4 marks)

Question 44

The diagram shows a sketch of part of the curve C with equation $y = -x^2 + 2x + 4$.



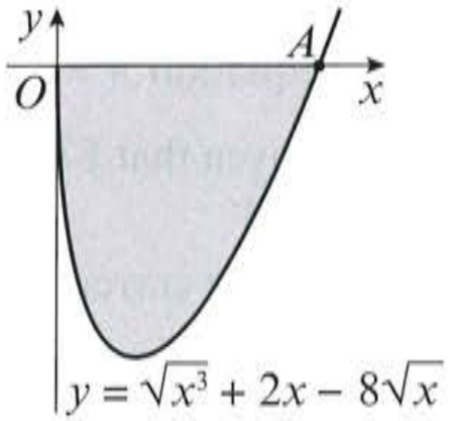
The straight line $y = x + 2$ crosses the curve at the points A and B.

a. Find the coordinates of points A and B. (2 marks)

b. The the area of the finite region R. (7 marks)

Question 53

The diagram shows part of the curve with equation $y = \sqrt{x^3} + 2x - 8\sqrt{x}$.



The curve intersects the x-axis at the origin and at the point A.

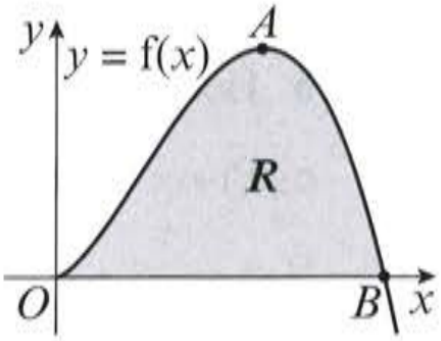
Find:

a. the x-coordinate of the point A (3 marks)

b. the area of the shaded region. (6 marks)

Question 79

$f(x) = 4x^{3/2} - \frac{1}{2}x^3$. The diagram shows part of a sketch of the curve with equation $y = f(x)$.



The shaded region R is bounded by the curve and the x-axis.

a. Find the coordinates of point A in the form $(\sqrt[3]{p}, q)$ where p and q are integers to be found. (3 marks)

b. Verify that point B has coordinates $(4, 0)$. (1 mark)

c. Find the exact area of the region R. (6 marks)

Answers

13.1 Integrating x^n

1.
- **a.** $3x^3 + c$
- **b.** $-6x^{-2} + c$
- **c.** $\frac{12}{7}x^{7/2} + c$
- **d.** $25x + c$

2.
- **a.** $\frac{2}{3}x^{3/2} + c$
- **b.** $\frac{9}{2}x^{2/3} + c$
- **c.** $4x^{5/4} + c$
- **d.** $-\frac{1}{3}x^{1/2} + c$

3.
- **a.** $\frac{1}{5}x^5 + \frac{1}{4}x^4 - \frac{5}{2}x^2 + 6x + c$
- **b.** $\frac{4}{3}x^3 - \frac{1}{3}x^{-1} + \frac{5}{2}x^{-2} + c$
- **c.** $4x^{3/4} - 8x^{1/4} + c$

4.
- **a.** $16x^2 + 24x + 9$
- **b.** $\frac{16}{3}x^3 + 12x^2 + 9x + c$

5. $3x^3 - 15x^2 + 25x + c$

6. $x - 3x^2 + 3x^3 + c$

7.
- **a.** 2
- **b.** $16x - 80x^2 + 200x^3 - 250x^4 + 125x^5 + c$

8.
- **a.** $16 - 16\sqrt{x} + 4x$
- **b.** $16x - \frac{32}{3}x^{3/2} + 2x^2 + c$

9.
- **a.** $6561 - 34992x + 81648x^2$
- **b.** $6561x - 17496x^2 + 27216x^3 + c$

13.2 Indefinite integrals

1.
- **a.** $\frac{1}{5}x^5 + c$
- **b.** $-x^7 + c$
- **c.** $-4x^{-1} + c$

2.
- **a.** $\frac{5}{3}x^3 - \frac{9}{2}x^{-2} + c$
- **b.** $\frac{1}{8}x^6 + x^{-4} + 4x^2 + c$
- **c.** $\frac{8}{5}x^{5/2} + 20x^{-1/2} + c$

3.
- **a.** $36x - 40x^{3/2} + \frac{25}{2}x^2 + c$
- **b.** $\frac{18}{5}x^{5/2} - 8x^{3/2} + 8x^{1/2} + c$
- **c.** $-5x^{-1} - \frac{7}{2}x^2 - \frac{8}{5}x^{-5/2} + c$

4.
- **a.** $2Px^{1/2} - \frac{2}{5}Rx^{5/2} + c$
- **b.** $-\frac{1}{2}Ax^{-2} - \frac{1}{4}Bx^{-4} + c$
- **c.** $\frac{3}{4}Mx^{4/3} - \frac{4}{7}Nx^{7/4} + c$

5. $\frac{180}{7}x^{7/5} - 50x^{6/5} + 25x + c$

6. $\frac{3}{4}x^4 - \frac{14}{3}x^{3/2} + 6x^{2/3} + 2x^{-1} + c$

7. $a = 3$, $b = 2$

8.
   a. $65536 - 262144x + 245760x^2$
   b. $65536x - 131072x^2 + 81920x^3 + c$

13.3 Finding functions

1.
   a. $y = x^4 - \frac{10}{3}x^3 + 15$
   b. $y = 10x^{1/2} + \frac{1}{6}x^3 - \frac{1973}{3}$
2.
   a. $y = -x^{-3} + x^{-1} + 4$
   b. $y = \frac{1}{3}x^3 - 5x^2 + 25x - \frac{44}{3}$
3. $16x^{1/2} - \frac{8}{3}x^{3/2} - 6$
4. $y = 10x - 3x^2 - \frac{4}{3}x^3 + 18$
5. $f(x) = -10x^{-1/2} - 6x^{1/2} + 24$
6. $y = 3x^2 - 6x^{2/3} - 56$
7.
   a. $p'(x) = \dfrac{1 - 15x + 75x^2 - 125x^3}{x^{3/2}}$
   $= x^{-3/2} - 15x^{-1/2} + 75x^{1/2} - 125x^{3/2}$
   b. $p(x) = -2x^{-1/2} - 30x^{1/2} + 50x^{3/2} - 50x^{5/2} + 8$
8. $-37 - \frac{1}{2}\sqrt{2}$

13.4 Definite integrals

1.
   a. $\frac{992}{3}$
   b. $\frac{135}{2}$
   c. $\frac{3}{8}$
   d. 10
2.
   a. 25.5
   b. 446
   c. 2385
3. $\frac{13}{8}$
4. $\frac{1359}{8}$
5. 161.5
6. 25
7.
   a. $y = (2x^{1/3} - \frac{1}{2}x^{-1/3})^2 = 4x^{2/3} + \frac{1}{4}x^{-2/3} - 2$
   b. $\frac{1223}{20}$
8. -9

13.5 Areas under curves

1. $\frac{4}{3}$
2.
   a. $(2, 0)$
   b. $\frac{45}{2}$
3. $\frac{75}{8}$
4.
   a. $A(-2, 0)$, $B(6, 0)$
   b. $\frac{256}{3}$
5. $\frac{27}{4}$
6.
   a. $A(4, 0)$
   b. $\frac{64}{3}$
7. 3

13.6 Areas under the x-axis

1. $\frac{125}{6}$
2.
   a. 2 and 4
   b. $\frac{16}{3}$
3.
   a. $\frac{8}{3}$
   b. $\frac{5}{12}$
   c. $\frac{37}{12}$
4.
   a. $(4, 0)$
   b. $\frac{64}{3}$
5.
   a. $A(3, 0)$, $B(10, 0)$
   b. $\frac{343}{12}$
6.
   a. $A(-4, 0)$, $B(1, 0)$
   b. $\frac{131}{4}$
7.
   a. $f(2) = 8 + 24 - 8 - 24 = 0$
   b. $f(x) = (x - 2)(x + 2)(x + 6)$
   c. 128

13.7 Areas between curves and lines

1.
   a. $A(-4, 6)$, $B(4, 6)$
   b. $\frac{80}{3}$
   c. 48
   d. $\frac{64}{3}$
2.
   a. $P(1, 3)$, $Q(4, 0)$
   b. 9
   c. $\frac{9}{2}$
   d. $\frac{9}{2}$
3.
   a. $\frac{554}{15}$
   b. $\frac{194}{15}$

4.
- **a.** $A(3,5)$, $B(7,3)$
- **b.** $\frac{16}{3}$

5. $\frac{31}{30}$

6.
- **a.** $5x + \frac{16}{x^2} - 21 = 0$, then multiply by x^2 to get $5x^3 - 21x^2 + 16 = 0$
- **b.** $(4, \frac{1}{4})$
- **c.** $\frac{27}{8}$

7.
- **a.** $-18 - 9x = -x^3 - 2x^2 + 16x + 32$
  $\Rightarrow x^3 + 2x^2 - 25x - 50 = 0$
- **b.** $(5, -21)$
- **c.** $\frac{4459}{36}$ or 123.86 (2 d.p.)

Problem-solving Set A

Bronze

- **a.** $f'(x) = -5x^{3/2} + 7x^{1/6}$
- **b.** $f(x) = -2x^{5/2} + 6x^{7/6} - 14$

Silver 113.3

Gold $f(x) = x^3 - \frac{5}{2}x^2 - 8x^{-1} + 17$

Problem-solving Set B

Bronze

- **a.** $A(-4,9)$, $B(4,9)$
- **b.** $\frac{128}{3}$

Silver $\frac{32}{3}$

Gold

- **a.** $3x + 6 = -x^3 - 2x^2 + 4x + 8 \Rightarrow x^3 + 2x^2 - x - 2 = 0$
- **b.** $\frac{37}{12}$

Selected exam question bank answers

Question 6 $\frac{27}{5}x^{5/3} - \frac{4}{x} - \frac{3}{4}x^4 + 2x + c$

Question 8 $\frac{203}{64}$

Question 26 25

Question 44

a. $A(-1,1)$, $B(2,4)$
b. $\frac{9}{2}$

Question 53

a. 4
b. $\frac{208}{15}$

Question 79

a. $(\sqrt[3]{16}, 8)$
b. $f(4) = 4 \times 8 - \frac{1}{2} \times 64 = 0$
c. $\frac{96}{5}$